

RobustSTL: A Robust Seasonal-Trend Decomposition Algorithm for Long Time Series

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ABSTRACT

Decomposing complex time series into trend, seasonality, and remainder components is an important task for time series anomaly detection and forecasting. Although numerous methods have been proposed, many real-world time series characteristics remain difficult to address, including seasonality fluctuation and shift, abrupt change in trend and remainder, robustness to anomalies, and applicability to long seasonal periods. This paper proposes a novel and generic time series decomposition algorithm to address these challenges. The trend component is extracted robustly by solving a regression problem with least absolute deviations loss and sparse regularization. Based on the extracted trend, non-local seasonal filtering is applied to extract the seasonality component. This process is repeated until an accurate decomposition is obtained. Experiments on synthetic and real-world datasets demonstrate that RobustSTL outperforms existing solutions.

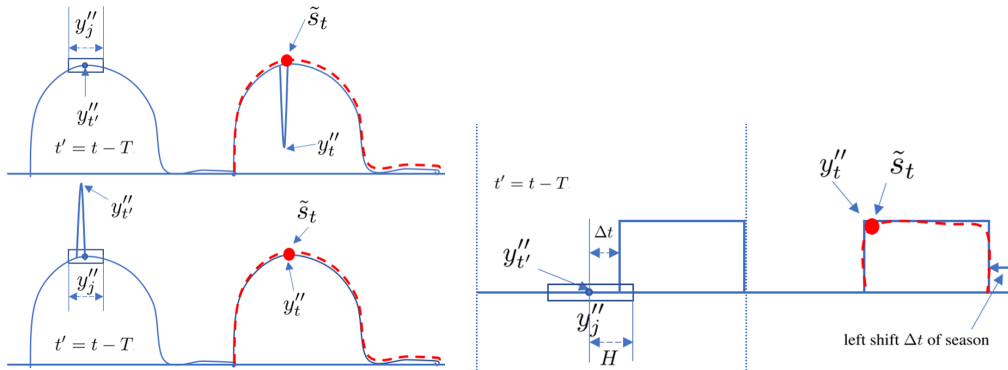


Figure 1: Overview of RobustSTL.